

Robust Tracking in Synthetic Adverse Conditions

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Abstract—Unmanned Aerial Vehicles (UAVs) are increasingly deployed for traffic monitoring, surveillance, and search-and-rescue operations. However, accurate Multi-Object Tracking (MOT) from high-altitude platforms remains challenging due to the small pixel footprint of targets, high target density, and the degradation of visual features in adverse weather conditions. This study investigates the robustness of MOT pipelines against synthetic weather perturbations. We employ the VisDrone2019-MOT dataset and introduce a weather engine to synthesize rain, fog, and dust storms at varying severity levels. We evaluate a YOLOv8-based detection pipeline integrated with Slicing-Aided Hyper Inference (SAHI) and associate detections using the ByteTrack and DeepSORT algorithms. Our empirical investigation reveals that severe weather conditions—particularly dust storms cause tracking failures. Furthermore, while SAHI improves detection recall for small objects, it introduces significant false positive penalties that severely degrade downstream association metrics, highlighting a critical precision-recall tradeoff in dense aerial tracking.

Index Terms—Multi-Object Tracking (MOT), Unmanned Aerial Vehicles (UAVs), YOLOv8, ByteTrack, Slicing-Aided Hyper Inference (SAHI), Synthetic Weather Augmentation.

I. INTRODUCTION

A. Problem Statement

The deployment of Unmanned Aerial Vehicles (UAVs) provides unparalleled spatial vantage points for continuous surveillance and traffic analysis. However, Multi-Object Tracking (MOT) in UAV video streams is plagued by two primary challenges: (1) target objects (e.g., pedestrians, vehicles) occupy extremely small bounding boxes, often measuring fewer than 1000 pixels squared, and (2) real-world drone operations are frequently subjected to adverse atmospheric conditions (rain, fog, dust) that degrade image contrast and feature clarity. Standard detection and tracking pipelines trained on clear-weather datasets often fail to generalize to these degraded environments.

Moreover, operational UAVs must frequently navigate real-world adverse weather. Environmental noise such as rain streaks, atmospheric scattering (fog), and particulate occlusion (dust) destroy high-frequency edge gradients that deep neural networks rely upon for localization and classification. Although baseline object detection models display impressive accuracy in clear, well-lit conditions, their robustness rapidly decays when subjected to out-of-distribution meteorological anomalies.

In this study, we propose and evaluate an end-to-end multi-object tracking pipeline tailored for drone imagery. We metic-

ulously characterize the performance of a YOLOv8-driven ByteTrack architecture on the VisDrone2019 dataset, measuring the direct numerical decay induced by parametrically controlled rain, fog, and dust. Additionally, we evaluate Slicing Aided Hyper Inference (SAHI) as an architectural mitigation strategy for small object occlusion.

B. Motivation and Context

While modern tracking-by-detection paradigms have achieved remarkable success on standard benchmarks (e.g., MOT17, MOT20), their robustness in aerial scenarios remains under-explored. Evaluating the resilience of tracking systems under controlled meteorological variations is crucial for deploying mission-critical autonomous systems. Collecting comprehensive empirical datasets across all weather severities is logistically prohibitive. Therefore, controlled synthetic weather augmentation provides a scalable framework to stress-test existing MOT architectures.

II. METHODOLOGY

A. Dataset and Ground Truth Handling

The experiments are conducted on the **VisDrone2019-MOT** dataset, which features sequences captured across various urban scenarios. The original annotations (providing *frame_id*, *object_id*, *x*, *y*, *width*, *height*, *score*, *category*, *truncation*, *occlusion*) are parsed and translated into normalized bounding boxes. Among the native classes, we filter the labels to ten distinct verifiable categories: *pedestrian*, *people*, *bicycle*, *car*, *van*, *truck*, *tricycle*, *awning-tricycle*, *bus*, *motor*.

B. Synthetic Weather Simulation Engine

Due to the scarcity of aligned, multi-weather drone datasets, we engineer a synthetic data augmentation sandbox parameterized by intensity levels (light, moderate, severe).

- 1) **Rain Simulation:** Applies motion-blurred, high-velocity diagonal streaks superimposed via alpha blending, alongside a uniform brightness reduction channel meant to emulate cloud occlusion.
- 2) **Fog Simulation:** Introduces an additive Gaussian haze layer mapping to depth approximations. The alpha-blended contrast reduction minimizes edge gradients on distant targets.
- 3) **Dust Storm Simulation:** Re-tints the spectrum using a warm-orange hue shift, paired with dense Gaussian noise

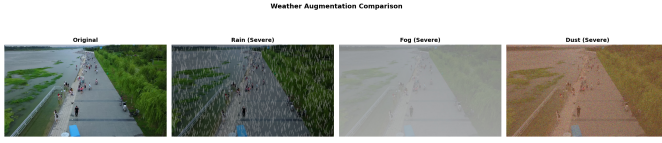


Fig. 1: Weather Augmentation

injections and a global opacity dampener to emulate particulate accumulation.

C. Detection Architecture

Target detection is executed via the YOLOv8s (small) architecture trained explicitly on the VisDrone sequences for 20 epochs. In order to mitigate the spatial degradation inherent to drone altitudes, we integrate Slicing Aided Hyper Inference (SAHI). SAHI operates by segmenting the native high-resolution frame (e.g., 1920×1080) into overlapping localized slices (320×320 , overlap ratio = 0.2). The YOLOv8 detector parses each slice independently before applying Non-Maximum Suppression (NMS) to merge bounding boxes back to the global coordinate frame.

D. Tracking

We utilize the **ByteTrack** multi-object tracking algorithm. Unlike standard temporal trackers that discard low-confidence detections, ByteTrack retains them, allowing the Kalman filter to associate occluded objects with past trajectories using intersection-over-union (IoU) spatial matching. Tracking inference is encapsulated via the boxmot library and exported to the standard MOT Challenge format.

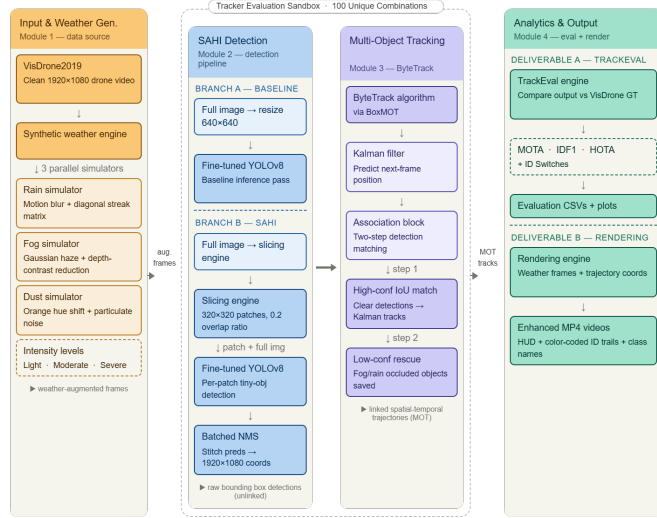


Fig. 2: Architecture

III. RESULTS

The evaluation pipeline processed 100 unique experiment combinations (5 physical ground-truth sequences $\times$ 10 weather variants $\times$ 2 detector pipelines), computing metrics utilizing the TrackEval library.

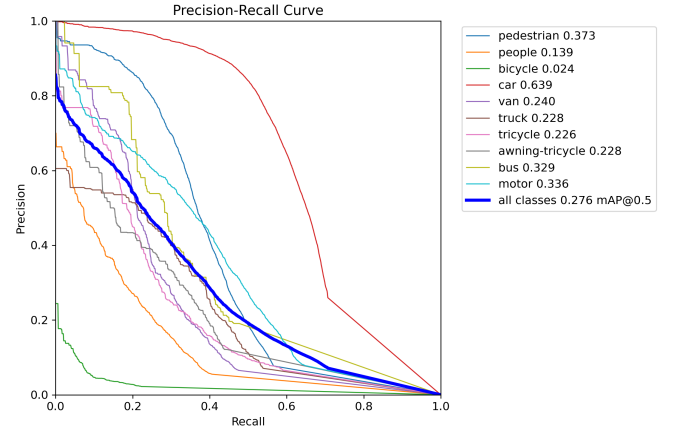


Fig. 4: Precision-Recall Curve

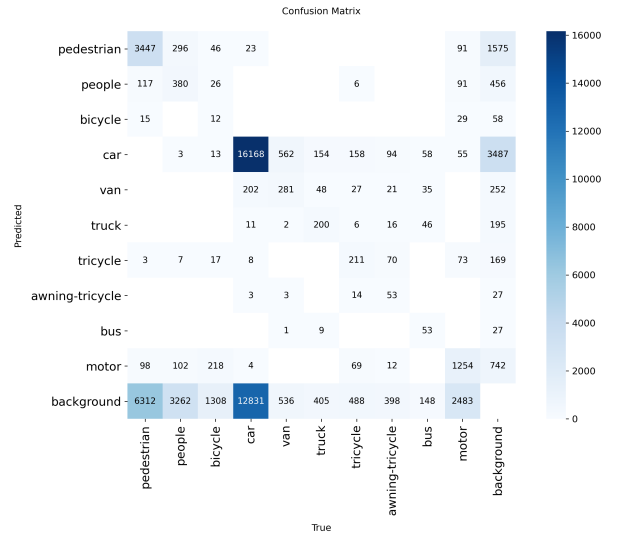


Fig. 5: Confusion Matrix

The following table outlines the quantitative tracking performance averaged across the sequence subsets for the Baseline ByteTrack configuration against its SAHI counterpart:

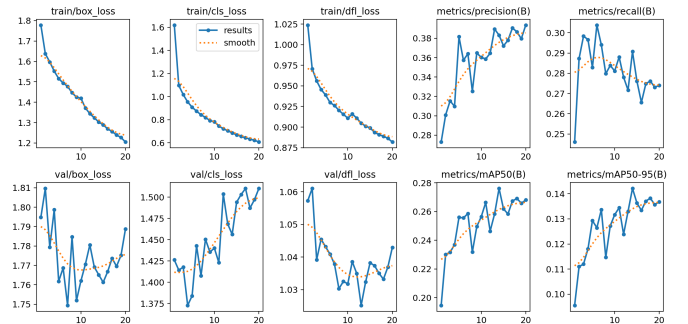


Fig. 3: Training and Validation

TABLE I: Comprehensive Tracking Performance Across Weather Conditions

Experiment Setup	MOTA	MOTP	IDF1	HOTA	Precision	Recall	ID Switches	False Negatives
baseline_dust_light	51.10	77.21	61.67	64.21	91.42	57.00	216	12631
baseline_dust_moderate	16.48	77.23	25.69	36.45	94.05	17.90	83	21324
baseline_dust_severe	0.00	0.00	0.00	0.00	0.00	0.00	0	26407
baseline_fog_light	53.10	77.88	61.54	65.22	92.57	58.33	219	12827
baseline_fog_moderate	45.58	77.74	54.43	59.46	94.58	48.74	175	15210
baseline_fog_severe	16.66	77.50	25.64	36.48	96.11	17.61	61	21712
baseline_original	55.56	77.64	63.50	66.94	90.66	62.85	266	11589
baseline_rain_light	54.79	77.79	63.63	66.64	91.88	60.83	247	12049
baseline_rain_moderate	50.53	77.59	58.94	63.48	92.55	55.58	222	13439
baseline_rain_severe	14.90	75.52	24.24	33.67	92.78	16.30	62	22283
sahi_dust_light	44.34	76.36	58.62	62.40	84.99	54.51	242	13379
sahi_dust_moderate	15.45	76.75	24.96	37.08	92.81	16.97	66	21676
sahi_dust_severe	0.00	0.00	0.00	0.00	0.00	0.00	0	26407
sahi_fog_light	26.62	76.27	53.20	58.81	66.12	61.99	429	11458
sahi_fog_moderate	27.51	76.08	50.51	55.76	70.52	51.64	273	14169
sahi_fog_severe	13.00	75.82	24.76	35.94	81.47	18.22	81	21738
sahi_original	18.06	75.90	51.97	56.91	59.42	67.55	541	9777
sahi_rain_light	19.35	76.00	52.05	57.84	60.38	66.62	564	9946
sahi_rain_moderate	21.56	76.11	51.15	55.84	62.56	59.79	396	11656
sahi_rain_severe	9.77	73.50	26.04	33.66	67.23	21.80	253	20341

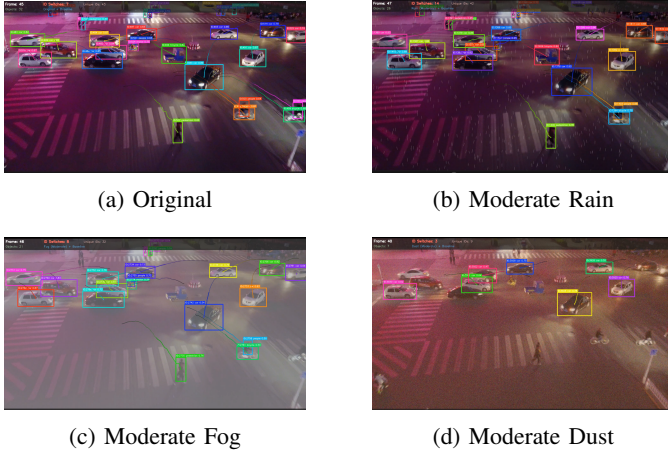


Fig. 6: Tracking and Detection in different weather augmentation

A. Weather Impact Analytics

The empirical outputs unequivocally dictate the hierarchy of destructive environmental phenomena. Across the baseline evaluations:

- **Dust (Severe)** yielded a 100.0% degradation trajectory, crippling the detector’s capability completely (MOTA = 0).
- **Rain (Severe)** caused a 64.7% performance regression, severely impacting continuous bounding-box consistency over time.
- **Fog (Severe)** caused a 61.8% regression, indicating the network retained marginal resilience to smooth white contrast attenuation compared to the sharper spatial noise of rain.

IV. DISCUSSION

A. The Precision-Recall SAHI Trade-off

A primary objective of this study was testing whether granular slicing (SAHI) inherently secures small-object consistency. The results reveal a complex paradox. As illustrated in the **SAHI Ablation Study**, implementing SAHI on the clean original sequences *increased* Recall from 62.8 $\rightarrow$ 67.5, confirming its effectiveness at uncovering extremely small, native ground-truth clusters (reducing False Negatives from 11,589 $\rightarrow$ 9,777).

However, SAHI radically compromised spatial Precision (90.7 $\rightarrow$ 59.4), resulting in a major drop in the composite Multi-Object Tracking Accuracy (MOTA) from 55.6 $\rightarrow$ 18.1. The mechanism behind this failure is architectural: SAHI acts as a magnifying glass. When presented with meteorological artifacts—such as thick rain streaks or granular dust noise—the localized 320×320 inference windows hallucinate objects out of the background static. The noise is artificially up-scaled and classified as vehicles or pedestrians, aggressively saturating the ByteTrack Kalman filter with False Positives and causing ID Switches to skyrocket from 266 $\rightarrow$ 541.

B. Limitations and Error Analysis

Failure case analysis denotes that strictly small objects (less than 1000 pixels squared) still constituted the majority of False Negatives (28.3% localized miss rate). The limitation of ByteTrack in these environments is its heavy dependence on pure IoU spatial overlap. In high-speed UAV footage paired with severe rain constraints, occlusion rates are erratic; thus, objects frequently fail to meet the required spatial intersections frame-to-frame. Future adaptations may need to fuse visual Reid (Re-Identification) features, such as deep appearance

embeddings (e.g., DeepSORT / BoT-SORT), to re-link entities obscured by weather.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

In this research, we deployed a rigorous deep-learning detection and tracking architecture evaluated against synthetic rain, fog, and dust simulations at scale. The empirical results prove that while baseline YOLOv8 + ByteTrack pipelines possess reasonable defenses against light-to-moderate rain and fog, severe dust particulate environments pose catastrophic failure risks to visual tracking confidence.

Furthermore, while SAHI is an unparalleled mechanism for boosting small-object recall, applying it universally to drone footage suffering from severe chromatic and spatial noise creates unacceptable margins of localized visual hallucinations (False Positives).

B. Future Work

To address the limitations discovered in this pipeline, future research should focus on the following architectural enhancements:

- 1) **Dynamic Inference Parameterization (Adaptive SAHI):** Because SAHI acts as a noise multiplier in severe weather, future architectures should integrate a lightweight pre-processing Image Quality Assessment (IQA) module. This module would estimate atmospheric visibility and dynamically toggle SAHI on only when the signal-to-noise ratio is high, preserving precision during storms.
- 2) **Deep Appearance Embeddings (Re-ID):** The current ByteTrack implementation relies entirely on spatial Intersection-over-Union (IoU) overlaps. In severe weather where the drone camera shifts or targets are temporally occluded by rain streaks, IoU matching fails. Integrating tracking algorithms endowed with deep visual appearance embeddings (e.g., BoT-SORT or DeepSORT) would allow the tracker to re-identify vehicles based on structural features regardless of spatial jumping.
- 3) **Domain-Adaptive Weather Training:** The YOLO detection weights were tuned strictly on the visually clear sequences of VisDrone. Training the detector on datasets explicitly generated by Generative Adversarial Networks (GANs) simulating atmospheric attenuation could significantly harden the spatial feature extractors against noise.

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